

Project report



Source: [Grutto. Deer grazing in the Oostvaardersplassen Nature Reserve.](#)

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Abstract

This report looks at the impact of introducing grey wolves into the Oostvaardersplassen Nature Reserve in the Netherlands. The reserve has been struggling with overpopulation of herbivores like Red Deer, which is damaging vegetation and disrupting the ecosystem. Using a mathematical model, we simulated what would happen if wolves were added to the area. This approach has been implemented in a Java application, allowing users to explore different scenarios and predict ecological outcomes. The results show that wolves can help bring the Red Deer population down to a healthier level, which in turn reduces competition for resources among other herbivores like Konik horses and Heck cattle. The study also discusses possible risks, like over-predation, and ethical concerns, such as how these changes might affect animal welfare and public opinion. Overall, this project shows how wolves could help restore balance to the reserve and offers a tool for managers to make better decisions.

1. Introduction

The Oostvaardersplassen Nature Reserve, located in Flevoland, the Netherlands, is a remarkable ecological area that is home to various herbivores like wild horses, deer, and cattle. However, the lack of apex predators has created an ecological imbalance. Without predators to regulate their populations, herbivores have grown unchecked, leading to vegetation damage and overall instability within the ecosystem. This situation presents significant challenges for the reserve's management, which seeks to preserve natural processes with minimal human intervention.

To address these issues, the project aims to create a mathematical modeling and simulation tool to explore the potential impacts of introducing a top predator, specifically grey wolves, into the ecosystem. By modeling predator-prey dynamics and interspecies interactions, this tool will provide valuable predictions about ecological outcomes, helping reserve managers make more informed decisions.

The foundation of the simulation relies on established mathematical models, such as exponential and logistic growth equations, Lotka-Volterra predator-prey dynamics, and interspecies competition frameworks. These models are implemented in a Java-based application, using computational techniques to simulate how populations change over time. The project also incorporates environmental data, animal behavior patterns, and visualization tools to create an interactive platform that allows users to explore different ecological scenarios.

This report provides an in-depth look at the mathematical models and methods used in the project, explaining both their theoretical background and practical applications. Ultimately, the goal is to support sustainable ecosystem management by offering a reliable, data-driven decision-making tool for the Oostvaardersplassen Nature Reserve.

1.1 Context

The Oostvaardersplassen Nature Reserve covers 5,600 hectares, with 3,600 hectares of wetlands and 2,000 hectares of dry land. Created as a natural sanctuary, the reserve is home to a variety of herbivores, including wild horses, cattle, and deer, as well as geese and birds of prey. While its management focuses on preserving natural processes with minimal human interference, the lack of apex predators has disrupted the delicate ecological balance. The fenced boundary of the reserve further complicates the situation by preventing herbivores from migrating, which worsens overpopulation and leads to habitat degradation.

Current ecological patterns highlight the challenges posed by unchecked herbivore populations. For example, rising temperatures are expected to benefit wild horses but threaten the survival of wild cattle. While deer have a relatively minor impact on other herbivores, geese are a significant problem, as they heavily disrupt vegetation and compete for resources. Past attempts to control geese populations by introducing foxes led to unpredictable outcomes, illustrating how complex ecological interventions can be.

To address these challenges, introducing grey wolves has been proposed as a potential solution. As apex predators, wolves could naturally regulate herbivore populations, reduce overgrazing, and promote the recovery of vegetation. However, the broader ecological impacts of such an intervention remain uncertain. Striking a balance between the reserve's ecological needs, public perception, and ethical concerns requires advanced tools capable of predicting outcomes across different scenarios.

This project aims to meet this need by using mathematical modeling and simulation techniques to assess the potential effects of introducing predators. The simulation tool is designed to help reserve managers analyze population dynamics, evaluate risks, and consider ethical implications. By providing actionable

insights, the tool will support informed and sustainable management decisions for the Oostvaardersplassen Nature Reserve.

1.2 Main research question and subquestions

Main Research Question:

What will happen to the populations of deer, cattle and horses if we release a number of grey wolves in the preserve?

Sub-questions:

1. What are the current population dynamics of wild horses, cattle, deer in Nature Reserve, and how does that affect ecosystem?
2. What are the potential behavioral changes in herbivore populations following the introduction of apex predators?
3. What potential risks might the introduction of grey wolves into the reserve entail?
4. What are the limitations of the proposed mathematical model and simulation tool?
5. What ethical considerations should be considered when introducing top predators?

2. Methodology

2.1. Research Strategy

Our project will employ a quantitative, predictive research strategy. By developing a mathematical model and using simulation techniques, the project will quantitatively predict the ecological impacts of introducing gray wolves into the reserve. The model will draw on available ecological data to simulate different scenarios. The approach will be both exploratory (exploring the potential effects of introducing top predators) and predictive (predicting how the reserve ecosystem might respond over time).

The validity and reliability of the model will be ensured through a rigorous validation process comparing the model results with real data and expert opinion.

2.2. Research Design

Each sub-question will be addressed using specific methods:

1. Current Population Dynamics (Sub-question 1):

- Data Collection: Use historical and current population data from Oostvaardersplassen.
- Analysis: Develop a mathematical model to simulate current population trends and their ecological consequences.

2. Changes in Herbivore Behavior (Sub-question 2):

- Case Studies: Review studies documenting herbivore behavioral responses to the presence of predators.

3. Implementation Risks (Sub-question 3):

- Scenario Analysis: Identify and model potential risk scenarios

4. Model Limitations (Sub-question 4):

- Technical Review: Critically evaluate the assumptions, simplifications, and limitations of the mathematical model and modeling tool developed.
- Comparative study: compare the developed model with alternative ecological modelling tools.

5. Qualitative research Design (Sub-question 5):

- Literature Review: Analyze existing research, ethical frameworks, and case studies on predator reintroduction to identify key ethical themes.
- Case Study Analysis: Review 3-5 reintroduction projects (e.g., wolves in Yellowstone) to examine how ethical considerations were handled.

3. Results and analysis

Sub question 1:

- Current Dynamics of Herbivore Populations and Their Impact on the Ecosystem
In the absence of apex predators, the populations of wild horses, cattle, and deer in the Oostvaardersplassen ecosystem exhibit a predictable but ecologically harmful growth pattern. Initially, these populations experience a phase of exponential growth, driven by abundant resources and a lack of predation. This exponential growth can be described using the equation:

$$N_t = N_0 \cdot e^{r \cdot t}$$

Where:

$N(t)$ is the population size at time t , N_0 is the initial population size, r is the intrinsic growth rate ($r = b - d$), the difference between birth and death rates, and t is time. During this period, birth rates significantly exceed death rates, leading to rapid population increases. While this appears as a thriving ecosystem in the short term, it quickly sets the stage for resource depletion (Lotka, 1925[6]).

Over the past 30 years, we have seen an exponential growth in the red deer population, which peaked in 2019 at 3,250 individuals, after which the growth stopped, and the population reached its maximum of around 3,200 individuals.

A similar picture is also observed with the Konik horses but less rapid with an increase of 6150% compared to the red deer (7122%) with a maximum of 1250 individuals.

For the Heck cattle the picture is different having reached a peak of 680 (2167% of the original population) in 2012 the population began to decline sharply against the background of a rapid growth of

populations of other species that turned out to be more adapted to such an environment having exceeded the estimated carrying capacity by 1.6-2 times

Our model predicts that over the next few years the population will gradually decrease. Such a large excess of Carrying Capacity should lead to a further reduction in available grass.

Without the introduction of wolves, model predicts the following equilibrium levels for key species in the reserve:

- **Red Deer:** Population will reach a plateau at approximately **1800**.
- **Konik Horses:** Population will reach a plateau at approximately **700**.
- **Heck Cattle:** Population will reach a plateau at approximately **120**.

Despite these equilibrium levels, the **desired number of red deer** (500) will not be achieved naturally under current conditions.

Sub question 2: What are the potential behavioral changes in herbivore populations following the introduction of apex predators?

The introduction of wolves into the Oostvaardersplassen has the potential to induce several behavioral changes in the herbivore populations, particularly the Red Deer. Wolves, as apex predators, will not only reduce the deer population but also alter the behavior of surviving individuals. According to the concept of ecosystem-based carrying capacity from the Report of the second International Commission on Management of the Oostvaardersplassen (ICMO2, 2010) [\[5\]](#), the population size of herbivores must be maintained in such a way that it does not lead to excessive impacts on the ecosystem. In other words, herbivores should not reach a population size that causes overgrazing, which could lead to long-term degradation of the habitat.

In Yellowstone, the reintroduction of wolves caused significant changes in elk behavior, including altered foraging habits and reduced pregnancy rates (Fields, 2017) [\[3\]](#). These findings suggest that similar behavioral changes could occur in the herbivores in Oostvaardersplassen, particularly in Red Deer, following the introduction of grey wolves

Ecosystem and Predation Assumptions

Grey wolves primarily hunt Red Deer, so our calculations focused on their direct impact on the deer population. However, as opportunistic predators, wolves may also hunt Konik horses and Heck cattle, especially if deer become scarce. Here's how we factored these species into our calculations:

- **Deer Per Wolf Factor:** 250 deer per year per wolf.
- **Horse Per Wolf Factor:** 1000 horses per year per wolf, recognizing they are less common prey.
- **Cattle Per Wolf Factor:** 800 cattle per year per wolf, considering a less frequent predation pattern.

The average weights of these animals are:

- **Red Deer:** 80-150 kg.
- **Konik Horses:** 250-400 kg.
- **Heck Cattle:** 400-600 kg.

This approach allows to assign a "weight" or "attractiveness" to the more probable prey, also accounting for each animal's average weight. Our solution to the problem that Red Deer are a more frequent food source for grey wolves involves increasing the number of individuals needed to satisfy the wolves' primary needs, in proportion to the prey's weight. In other words, in relation to the current population of a particular species, the higher the number, the "less interesting" the species as prey and the less wolves will be attracted to it

"According to the International Wolf Center, a single grey wolf requires the equivalent of **15-20 deer per year** to meet its dietary needs."

However, our model used a higher value of **250 deer per wolf per year** to ensure the simulated carrying capacity effectively adjusts for interspecies dynamics and balances long-term ecosystem stability. This approach accounts for potential alternative food sources and maintains a safety margin on the calculated threshold. The larger number was chosen in proportion to other high values for different animals, recognizing that in an actual scenario, 20 deer per year might suffice for a wolf.

The study area covers around 5,600 hectares (56 km²), with 3,600 hectares of wetlands and 2,000 hectares of dry land, which is available for herbivores and predators.

Considering space limitations and ecological balance, a wolf pack size of 9-13 **wolves** was deemed appropriate for introduction.

Modeling the Dynamics

The Lotka–Volterra Predator-Prey Model, taken from *A Primer of Ecology* by Nicholas Gotelli [4], was used to simulate the interactions between wolves and herbivores, with adjustments made to fit the specific conditions of the system.

1. **Red Deer Growth:**
 - a. The Red Deer population (**N**) grows at an intrinsic rate (**r_{deer}**), but this growth slows as the population approaches the carrying capacity (**K_{deer}**) of the ecosystem.
2. **Competition Among Herbivores:**
 - a. Red Deer compete for resources with Konik horses (**H**) and Heck cattle (**C**), modeled using competition coefficients (**a_{dh}** and **a_{dc}**) to quantify their impact.
 - b. The equation integrates interspecific competition, ensuring the combined effect of all herbivores is reflected in the resource limitations.
3. **Predation by Wolves:**
 - a. Wolves (**P**) reduce the Red Deer population directly through the term **$-a*N*P$** , where **a** is the predation rate. This term ensures that predation increases as wolf numbers rise or as Red Deer populations grow.

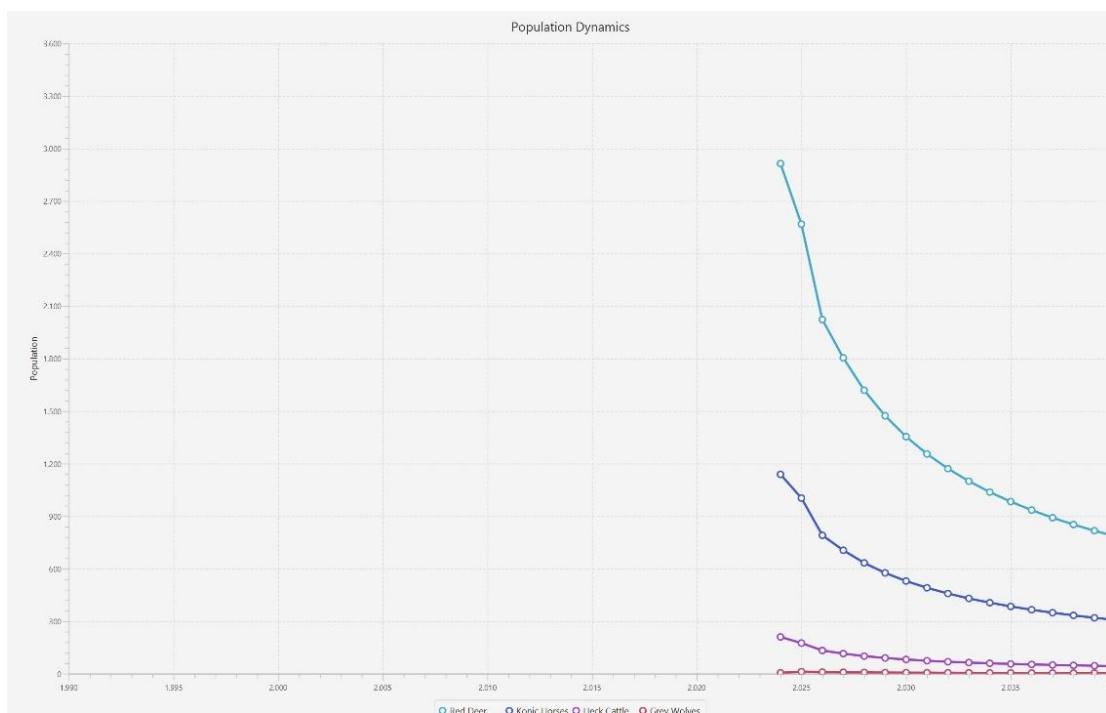
This term is however added only in the calculation of the Red Deer dynamics because this population is their primary prey. For the rest of the animals the formula stays the same, without the term of $-a \cdot N \cdot P$.

The full model for Red Deer dynamics is represented as:

$$\frac{dN}{dt} = r_{Deer} \cdot N \cdot \left(1 - \frac{N}{K_{Deer}} - \frac{\alpha_{DH} \cdot H}{K_{Horse}} - \frac{\alpha_{DC} \cdot C}{K_{Cattle}} \right) - a \cdot N \cdot P$$

Formula 1: Remodeled Lotka- Volterra Predator-Prey Model

Results Overview



Graph 1: Population trends for Red Deer, Konik horses, Heck cattle, and wolves from 2025 to 2040

The simulation results, presented in **Graph 1**, show the population trends for each species:

- Red deer: the population will decrease to around 830, much closer to the desired level of 500.
- Horse: the population will stabilize at around 300.
- Heck cattle: the population will stabilize at around 55.

Analysis

In the second year after the introduction of grey wolves, the Red Deer population experiences a sharp decline. This dramatic drop isn't because the wolves eat that many deer, but rather because of their hunting behavior. When prey is abundant, wolves may kill more than they need to consume, a phenomenon known as "surplus killing."

The high number of deer at the start of the experiment created an environment where wolves could easily hunt without limitation. This abundance led to a significant initial reduction in the deer population. However, as the number of deer decreased and stabilized closer to the ecosystem's carrying capacity, the wolves' predation naturally adjusted to match their actual dietary needs.

This highlights the important role of predators in balancing prey populations, particularly in cases where prey numbers initially exceed what the environment can sustain.

Introducing grey wolves has been a highly effective solution for managing the overpopulation of Red Deer. The combination of predation by wolves and interspecific competition among herbivores has shown how apex predators naturally regulate prey populations. This reduction in Red Deer numbers has also eased resource competition, indirectly benefiting Konik horses and Heck cattle, and promoting greater ecosystem balance. While wolves may occasionally prey on cattle or horses during times of prey scarcity, these events are rare, and the overall populations of these species remain stable under normal conditions.

Sub question 3: What potential risks might the introduction of grey wolves into the reserve entail?

The Oostvaardersplassen is a unique nature reserve in Flevoland, the Netherlands. Introducing grey wolves as a natural predator could be the solution to control the populations, but this might bring some risks to this ecosystem.

This might affect the behavior of the herbivores. Grey wolves could reduce the population of cattle, deer and horses. This could change the behavior, causing them to avoid open areas. Which would lead to uneven grazing pressure and could damage specific parts of the Oostvaardersplassen.

The Oostvaardersplassen is relatively very small and surrounded by human housing and infrastructure. Wolves probably will wander outside the reserve, leading to conflicts with farmers, which already is happening in the more eastern part of The Netherlands. There have been multiple attacks from wolves on sheep, dogs, chickens and some other animals in The Netherlands, like Gelderland, there has already been a sharp rise in wolf attacks on livestock, with 237 incidents reported in 2024 (DutchNews.nl, 2024) [2]. Other than that, the wolves in The Netherlands never really killed a person, there were only a few biting incidents on the news but none of them were in danger of their lives. So, for the farmers in that area, it would be a problem to get wolves introduced in their region, but when you look at the news about wolves in The Netherlands, they usually do not go to cities. There is no reason for them to leave the Oostvaardersplassen, because there is plenty of food already there.

There is also the public perception and opposition that would try to deny the wolf from being introduced. There are already some groups like the group: "No wolves Benelux" they talk about the wolves with the government and try to get rid of the wolf in the whole Benelux, because they think it is too crowded for a wolf to live here. Also, the damage and the costs that the wolves bring with them is in their eyes way too much.

The chance of encountering a wolf in the Netherlands is already very small. Wolves are shy and do not show themselves that easily. They primarily hunt during dusk and dawn. Wolves often hunt during the early hours of the morning when herbivores may be more vulnerable. They are also active during the evening, taking advantage of low light to stalk and get their prey.

Sub question 4: What are the limitations of the proposed mathematical model and simulation tool?

The proposed mathematical model and simulation tool's simplifying assumptions introduce some limitations. For example, the wolf population is kept constant for some years and is not allowed to vary with prey availability, territorial behaviors, or environmental conditions. This excludes factors such as surplus killing by wolves and the variation in predation patterns according to changing environmental conditions. Also, the model is focused on red deer dynamics and gives scant attention to the roles of Konik horses and Heck cattle, missing broader ecological implications.

On the other hand, the Grass Growth Simulation Model, which analyzes the relationship between grass height and temperature, is based on a linear regression equation, $y = mx + c$. Here, y represents grass height, x is the average temperature, m is the slope showing the change in grass height per degree, and c is the intercept or predicted height. This model examines historical data to predict grass heights for future temperatures using an average temperature. This method is widely recognized for its simplicity and effectiveness in establishing relationships between dependent and independent variables (Montgomery et al., 2012) [7]. A correlation analysis is also conducted by calculating the correlation coefficient using the formula: $r = \text{Cov}(X, Y) / \sigma_X \cdot \sigma_Y$, Where $\text{Cov}(X, Y)$ is the covariance between temperature and grass height. σ_X and σ_Y are the variances of the temperature and grass height, respectively. The negative correlation found clearly indicates that on days with a higher temperature, grass heights are decreased, showing how sensitive grass height is to variations in climatic conditions.

Linear though effective, this approach over-simplifies the grass growth processes, influenced by soil quality, precipitation, and herbivory. The model predictions are based on the extrapolation of historical data that may not consider fully the ongoing climate changes or other environmental dynamics.

While these models are instructive, careful interpretation is warranted. Findings from these models need to be supplemented with empirical observations, comprehensive ecological data, and expert evaluation to provide a robust and accurate understanding of decision-making.

Sub-Question 5: *What ethical considerations should be considered when introducing top predators?*

Introducing apex predators like grey wolves into ecosystems provides valuable ecological opportunities but also brings about complex ethical dilemmas. Predators are essential for maintaining balance by controlling herbivore populations, which in turn helps vegetation and other species thrive. However, such actions can lead to unforeseen outcomes, including increased suffering for prey, potential disruptions to ecosystems, and conflicts with humans. Responsible wildlife management must carefully weigh the benefits against the potential risks, striving to balance ecological restoration with respect for animal welfare and public concerns.

Minteer and Puschmann emphasize in their paper *The Ethics of Wildlife Management and Conservation: What Should We Try to Protect?* [1] that ethical considerations should highlight the intrinsic value of predators within ecosystems and their role in preserving biodiversity. This approach aligns with practical issues, such as reducing predation-induced suffering and preventing over-predation that could harm herbivore populations. The Prindle Institute for Ethics [8] also stresses the importance of gaining public understanding and support through transparent decision-making and active community involvement, especially for interventions that visibly affect animal populations.

Looking at Yellowstone National Park [9] as an example, the reintroduction of grey wolves successfully controlled excessive herbivore numbers, leading to vegetation recovery and greater biodiversity. Nevertheless, this initiative faced public resistance and raised ethical concerns, including how to balance

ecological benefits with issues like livestock losses and human safety, as highlighted by the U.S. National Park Service. Similar challenges would arise when introducing wolves into the Oostvaardersplassen Nature Reserve, making it essential to adopt measures such as compensation for affected farmers and fencing to reduce human-wildlife conflicts.

In conclusion, the introduction of grey wolves into the Oostvaardersplassen must tackle these ethical challenges by combining ecological objectives with societal values and animal welfare considerations. By learning from past experiences and employing transparent and adaptive management, this initiative can align with both conservation goals and community expectations.

4. Conclusion

The main research question was: *What are the ecological impacts and population dynamics resulting from the introduction of grey wolves into a restricted reserve ecosystem?*

To answer this, we looked at several subquestions:

1. What are the current population dynamics of wild horses, cattle, deer in Nature Reserve, and how does that affect ecosystem?
In the Oostvaardersplassen, the absence of apex predators has led to uncontrolled growth of herbivore populations. This imbalance has resulted in resource depletion and ecological strain. Despite equilibrium levels, the desired population of **500 red deer** won't be reached naturally. Thus, regulatory measures, such as introducing wolves and additional strategies, need to be implemented to restore the ecological balance.
2. What are the potential behavioral changes in herbivore populations following the introduction of apex predators?
As shown in Chapter 3, the introduction of grey wolves into the Oostvaardersplassen Nature Reserve leads to changes in herbivore behavior. Wolves directly reduce the Red Deer population, which lowers competition for resources and benefits Konik horses and Heck cattle. While wolves may occasionally prey on these species, their impact is minimal. Overall, the presence of wolves helps balance the ecosystem and improve resource availability.
3. What potential risks might the introduction of grey wolves into the reserve entail?
In Chapter 3, it's explained that introducing grey wolves to the Oostvaardersplassen could help manage herbivore populations, but it also brings risks. Wolves might alter herbivore behavior, causing uneven grazing and damage to the reserve. The area's small size and its closeness to human settlements increase the likelihood of wolves wandering out, leading to conflicts with farmers, as has already happened in other parts of the Netherlands. Although wolves rarely threaten humans, public concerns about livestock attacks, costs, and damages add to the challenges. Balancing these risks with the potential ecological benefits is a complex decision.
4. What are the limitations of the proposed mathematical model and simulation tool?
The mathematical model and simulation tool provide useful insights into predator-prey dynamics but have limitations. The assumption of a constant wolf population and a focus mainly on Red Deer

exclude important factors like surplus killing, territorial behavior, and broader impacts on Konik horses and Heck cattle. Similarly, the Grass Growth Simulation Model highlights the link between grass height and temperature but oversimplifies by ignoring factors like soil quality, precipitation, and herbivory.

5. What ethical considerations should be considered when introducing top predators?
Introducing grey wolves into the Oostvaardersplassen requires balancing societal values, ecological goals, and animal welfare through transparent and adaptive management.

Based on the answers to the subquestions, the main question can now be answered:

The introduction of grey wolves into the Oostvaardersplassen Nature Reserve has a profound impact on the ecosystem. As discussed in Chapter 3, wolves mainly affect the Red Deer population, causing it to decrease. This reduction in Red Deer numbers helps lower competition for resources, which ultimately benefits other herbivores like Konik horses and Heck cattle. While wolves do occasionally prey on these other species, their overall impact on them is less significant. The result is a more balanced ecosystem with better access to resources for all species involved.

However, bringing wolves into a relatively small, enclosed reserve does come with its risks. Wolves can change herbivore behavior, potentially causing uneven grazing patterns that could harm the vegetation. Additionally, since the reserve is close to human settlements, there's a chance that wolves might wander out of the area and come into conflict with local farmers, especially when it comes to livestock predation. The model used to predict these population dynamics also has its limitations, as it doesn't account for factors like surplus killing or territorial behavior, and the grass growth model overlooks environmental variables like soil quality and precipitation. Ethical concerns are another important consideration; introducing apex predators like wolves requires careful planning and an open dialogue with the public to minimize potential conflicts.

In the end, while introducing grey wolves to the Oostvaardersplassen could offer ecological benefits—helping to balance herbivore populations and improve resource availability—it's crucial that the process be carefully managed to handle both the risks to the ecosystem and the concerns of local communities.

5. Discussion

This study shows that introducing grey wolves into the Oostvaardersplassen Nature Reserve can help control the overpopulation of Red Deer and improve ecosystem balance. The results from the mathematical model suggest that wolves effectively reduce Red Deer numbers to sustainable levels, which benefits other herbivores like Konik horses and Heck cattle by decreasing competition for resources.

While the results are promising, there are some limitations to consider. The model assumes that the wolf population remains constant over time, which might not be realistic. In real life, wolf numbers could change depending on food availability, territory size, or environmental factors. The model also focuses mainly on Red Deer and gives less attention to other species, which might mean some ecological effects are not fully captured.

There are also risks associated with introducing wolves. For example, wolves might leave the reserve and come into conflict with nearby farmers, which has already been seen in other parts of the Netherlands.

Public opposition to wolves, especially from groups concerned about livestock or safety, is another important challenge to address.

Overall, this project shows that wolves can play a key role in managing ecosystems. However, it is important to continue monitoring the situation, improve the model to include more real-world variables, and involve local communities in planning and decision-making. This will help ensure that the introduction of wolves benefits the ecosystem while addressing the concerns of people who live near the reserve.

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Appendix A

To understand the model better, we should look at the mathematics behind it:

1. The Basic Lotka-Volterra Predator-Prey Equations

The classic Lotka-Volterra model is typically expressed as two coupled differential equations:

$$\frac{dN}{dt} = r \cdot N - a \cdot N \cdot P$$

$$\frac{dP}{dt} = b \cdot a \cdot N \cdot P - m \cdot P$$

Where:

- N : Prey population (e.g., Red Deer).
- P : Predator population (e.g., Grey Wolves).
- r : Prey's intrinsic growth rate.
- a : Predation rate (interaction between predators and prey)
- b : Conversion efficiency (rate at which prey biomass is converted into predator biomass).
- m : Predator mortality rate.

The second part, $-a \cdot N \cdot P$, represents the loss of prey due to predation. As the predator population increases, the predation pressure on the prey rises, leading to a decrease in the prey population.

2. Extended Lotka-Volterra Model for this Project

• Carrying Capacity:

The term $\left(1 - \frac{N}{K_{Deer}}\right)$ introduces logistic growth for the prey, ensuring the population stabilizes as it nears the carrying capacity of the environment (K_{deer}). This term models the density-dependent regulation of the prey population, which is crucial for realistic ecological modeling.

• Interspecific Competition:

Additional term $\frac{\alpha_{DH} \cdot H}{K_{Horse}}$ and $\frac{\alpha_{DC} \cdot C}{K_{Cattle}}$ account for competition with Konik horses (H) and Heck cattle (C), which limit resources available to Red Deer.

Putting everything together, the formula extended to this equation:

$$\frac{dN}{dt} = r_{Deer} \cdot N \cdot \left(1 - \frac{N}{K_{Deer}} - \frac{\alpha_{DH} \cdot H}{K_{Horse}} - \frac{\alpha_{DC} \cdot C}{K_{Cattle}}\right) - a \cdot N \cdot P$$

3. Linear Regression Formula (Grass Growth Simulation Model)

$$y = mx + c$$

where:

- y = height of the grass (cm)
- x = average yearly temperature (°C)
- m = Slope of the line (rate of change in grass height per degree)
- c = Intercept (grass height when temperature is zero)

4. Correlation Coefficient Formula (Quantify the relationship between grass height and temperature)

$$r = \text{Cov}(X, Y) / \sigma_X \cdot \sigma_Y$$

Where,

$\text{Cov}(X, Y)$ = Covariance of temperature and grass height

σ_X = standard deviation of temperature

σ_Y = standard deviation of heights

Appendix B

Grutto. *Deer grazing in the Oostvaardersplassen Nature Reserve.* <https://www.grutto.com/nl/onze-boeren/oostvaardersplassen>.

<https://figshare.com/articles/dataset/Data from the 26 predation studies performed in 17 different wolf territories in Scandinavia during 2001 2014 with year and length of study wolf pack size territory size 100 MCP moose and roe deer densities and number of moose and roe deer killed by wolve/4500239/1?file=7273487>

https://www.meteoblue.com/en/weather/historyclimate/climatemodelled/lelystad_the-netherlands_2751738
(Forecast)

<https://meteostat.net/en/place/nl/lelystad?s=06269&t> (Temperature)

<https://climateknowledgeportal.worldbank.org/country/netherlands/climate-data-historical> (Historical Data)

Appendix C

Application Screen

